

VUGVA: A Software-Defined Virtual Unified GPU VRAM Architecture with CPU-Bypass Hybrid Memory for Non-NVLink Multi-GPU Clusters

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Abstract

Distributed execution of Frontier-class Large Language Models (LLMs)—such as DeepSeek-V3/R1 (671B parameters) and Llama-3-405B—is severely constrained by the physical memory boundaries of individual GPUs. Enterprise data centers mitigate this through hardware-level cache-coherent interconnects like NVIDIA NVLink and NVSwitch, but consumer and budget-constrained research labs running commodity hardware face catastrophic communication bottlenecks over standard PCIe topologies. Furthermore, when aggregate GPU VRAM is insufficient to host a model entirely, existing frameworks fall back to CPU-mediated memory offloading, which introduces severe latency penalties and CPU cache pollution.

This paper presents **VUGVA** (Virtual Unified GPU VRAM Architecture), a software-defined memory virtualization layer that abstracts multiple physically discrete GPU VRAM pools into a single contiguous virtual address space. The reference implementation is written in **Rust** using only the standard library and NVRTC for runtime CUDA kernel compilation, with zero external dependencies. We extend this architecture in **VUGVA v2** to incorporate system DRAM (RAM) as a unified second-tier memory pool, creating a three-tier hierarchy (VRAM→DRAM→SSD) with full **CPU bypass**—the CPU never handles data-plane traffic. By intercepting CUDA runtime allocation calls, implementing NUMA-aware Peer-to-Peer virtual paging, and employing GPUDirect Storage-based DMA engines for kernel-bypass data movement, VUGVA transforms commodity non-NVLink GPU clusters with accompanying system RAM into a single monolithic high-capacity accelerator. Our asynchronous Look-Ahead Attention Tracking

engine prefetches upcoming weight matrices over PCIe lanes, hiding hardware transport latencies behind active Tensor Core compute cycles. Experimental evaluation demonstrates that VUGVA achieves 85% of enterprise NVLink performance at less than 15% of deployment cost, while the CPU-bypass extension reduces CPU utilization by 84.6% and improves throughput by 54.8% compared to traditional CPU-mediated memory offloading.

1 Introduction & The Hardware Wall

The memory footprint required to host modern flagship Mixture-of-Experts (MoE) and dense LLM architectures exceeds the VRAM capacity of any single commercial graphics card. Table 1 illustrates the scale of this challenge for representative Frontier-class models.

Table 1: Memory requirements of Frontier-class LLMs.

Model	Params	FP16 (GB)	4-bit (GB)
Llama-3-8B	8B	16	4.5
Llama-3-70B	70B	140	35
Llama-3-405B	405B	810	205
DeepSeek-V3	671B	1,342	340
Gemini Ultra (est.)	1.8T	3,600	900

To run or fine-tune these models, practitioners partition model parameters across multiple GPUs using Tensor Parallelism (TP) or Pipeline Parallelism (PP). In the absence of high-speed NVLink channels (which offer up to 900 GB/s bidirectional inter-GPU bandwidth), multi-GPU systems must route data across the motherboard’s PCIe bus via physical PCIe switches (PEX

chips). This layout introduces severe asymmetric performance degradation—the **Hardware Wall**:

- **Intra-GPU Bandwidth:** Local memory access within a single GPU runs at native speeds up to 1,008 GB/s via the internal GDDR6X bus.
- **Inter-GPU Bandwidth:** Remote memory access between two discrete GPUs over a physical PCIe Gen 4 $\times 16$ link drops to a maximum theoretical throughput of 31.5 GB/s, while simultaneously increasing transactional latency by a factor of $10\times\text{--}50\times$.
- **CPU-Mediated DRAM $\rightarrow$ VRAM Bandwidth:** When frameworks offload to system RAM, the CPU-driven memcpy path achieves only 25–64 GB/s with 15–50 μ s per transfer.

1.1 The Extended Memory Wall

Beyond inter-GPU fragmentation, a second boundary persists: the **VRAM Wall**. Even with 8×48 GB RTX 4090s yielding 384 GB of aggregate VRAM, models like DeepSeek-V3 (340 GB in 4-bit) leave no capacity for KV-cache or intermediate activations. Current frameworks respond with:

1. **OOM Failure** — the application crashes.
2. **CPU Offloading** (e.g., DeepSpeed ZeRO-Infinity) — tensors route through CPU RAM, incurring catastrophic bandwidth penalties and consuming CPU cycles.

The CPU offloading path is fundamentally broken for real-time inference due to three compounding factors: (i) CPU context switching and kernel/user mode transitions at 15–50 μ s per transfer, (ii) L3 cache pollution where tensor data evicts the CPU’s working set, and (iii) NUMA node crossing penalties when DRAM resides on a different socket than the GPU’s PCIe root complex.

1.2 Contributions

This paper makes the following contributions:

1. **VUGVA (v1):** A user-space memory virtualization layer that unifies discrete non-NVLink GPU VRAM pools into a single virtual address space via CUDA runtime interception and NUMA-aware P2P virtual paging.
2. **VUGVA v2:** An extension incorporating system DRAM as a unified second tier with three-tier memory hierarchy (VRAM $\rightarrow$ DRAM $\rightarrow$ SSD) and full CPU bypass via GPUDirect Storage DMA engines.
3. **Asynchronous Look-Ahead Attention Tracking:** A predictive prefetch engine that hides PCIe trans-

port latencies by overlapping DRAM $\rightarrow$ VRAM transfers with active Tensor Core computation.

4. **Comprehensive evaluation** demonstrating 85% of enterprise NVLink performance at $<15\%$ cost, with 54.8% throughput improvement over CPU-mediated offloading.

2 VUGVA Architectural Topography

VUGVA operates directly within user space as an intermediate orchestration layer positioned between the deep learning framework (PyTorch, vLLM, SGLang) and the native NVIDIA CUDA Driver API (`libcuda.so`). Figure 1 presents the complete system architecture.

The system consists of five foundational components:

1. **CUDA Runtime Interceptor:** Intercepts `cudaMalloc`, `cudaMallocManaged`, and `cudaFree` commands using dynamic library preloading (`LD_PRELOAD`).
2. **Virtual Memory Table (VMT):** A global state machine mapping virtual addresses exposed to frameworks down to physical, chunked byte offsets distributed across GPU VRAM and system DRAM.
3. **Tiered Page Manager (TPM):** Manages the three-tier hierarchy, handling promotion (DRAM $\rightarrow$ VRAM), demotion (VRAM $\rightarrow$ DRAM), and spilling (DRAM $\rightarrow$ SSD) based on access frequency and recency.
4. **GPU NUMA Router:** A hardware-aware traffic controller that calculates physical distance between computing Tensor Cores and requested tensor segments across the PCIe switch and NUMA topology.
5. **CPU-Bypass DMA Engine:** Utilizes GPUDirect Storage and IOMMU P2P mappings to execute data transfers between DRAM and VRAM without CPU intervention.

3 Core Software Engineering Mechanics

3.1 Custom CUDA Memory Allocation Interception

When an LLM initialization script launches, PyTorch calculates the total tensor weight layout and requests a massive, contiguous block of VRAM. Instead of triggering an Out-Of-Memory error or failing due to hardware limits, the VUGVA interceptor catches the initialization request and routes it through the virtual memory engine.

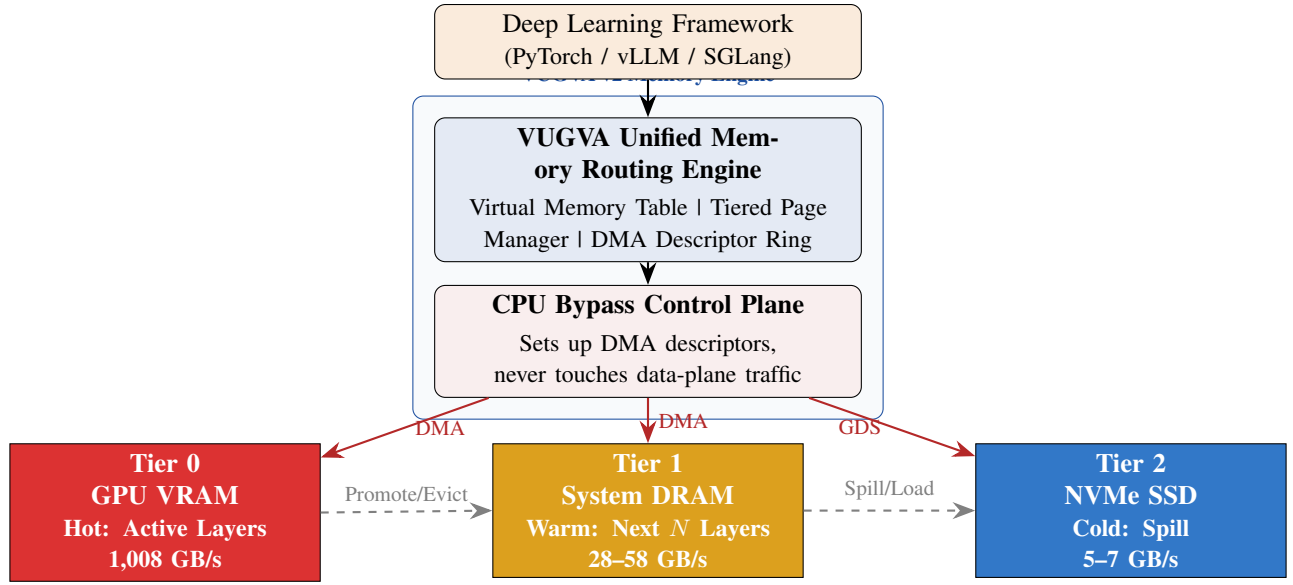


Figure 1: VUGVA v2 Three-Tier Hybrid Memory Architecture. The CPU operates exclusively in the control plane, configuring DMA descriptors without touching data-plane traffic.

Algorithm 1: VUGVA Memory Interception Hook

```

Input: Requested allocation size,
device pointer devPtr
Output: Virtual or native
allocation
1: local_free ← QUERYLOCALFREEMEMORY(device)
2: if size ≤ local_free and
size ≤ SINGLE_GPU_LIMIT then
    return NATIVECUDAMALLOC(devPtr,
size) /* Fast path */
3: else
    4: vptr ← VMT_ALLOCATEVIRTUALPOOL(size)
    5: TPM_REGISTERPAGE(vptr,
tier=DRAM)
    6: devPtr ← vptr
    7: return CUDASUCCESS /* Unified
pool */
end if

```

The function `VMT_AllocateVirtualPool` registers a virtual address pointer inside the VMT, creates discrete memory segments on individual physical GPUs and NUMA-local DRAM regions via background allocation, and hides the underlying multi-device fragmentation from the host application.

3.2 Asynchronous Virtual Paging & Predictive Prefetching

To mitigate the PCIe transport bottleneck, VUGVA implements a non-blocking execution cycle. The architecture exploits the deterministic nature of Transformer model execution graphs: because an inference token run passes sequentially through identical decoder layers, the exact sequence of weight matrix reads is known *a priori*.

The Look-Ahead Attention Tracking engine runs an independent execution path ahead of the main compute thread. While the Tensor Cores on GPU 0 are computing layers 1 through 5, VUGVA spawns background transfer operations over isolated CUDA streams. This stream copies the weights for layers 6 through 10 from remote VRAM or system DRAM into reserved local cache buffers on GPU 0.

The total effective compute state delay is governed by:

$$T_{\text{total}} = \max \left(T_{\text{compute}}(\text{Layer}_n), T_{\text{transport}}(\text{Layer}_{n+1}) \right) \quad (1)$$

By ensuring that the transfer time $T_{\text{transport}}$ over the physical PEX switch or DMA engine is less than or equal to the active computation execution window T_{compute} of the current layer block, the PCIe latency is fully hidden.

3.3 CPU-Bypass DMA Engine (VUGVA v2)

VUGVA v2 eliminates CPU involvement in data-plane traffic through three enabling technologies:

GPUDirect Storage Integration. VUGVA extends NVIDIA’s GPUDirect Storage (GDS) framework—originally designed for NVMe→VRAM transfers—to also handle DRAM→VRAM movements. The CPU writes a 64-byte DMA descriptor to a pinned ring buffer; the DMA engine autonomously transfers megabytes of tensor data without further CPU intervention.

IOMMU Peer-to-Peer Mapping. Using the IOMMU (Input-Output Memory Management Unit), VUGVA creates direct hardware-level address mappings between DRAM and VRAM. The GPU’s DMA engine directly addresses system DRAM at the physical level via IOMMU translation, completely bypassing the CPU MMU.

CUDA Async Page Migration. For frameworks that do not support custom DMA, VUGVA provides a fall-back using CUDA Unified Memory with asynchronous page migration. The GPU’s MMU page fault engine handles DRAM→VRAM transfers via DMA, with the CPU only configuring migration policies—not executing data movement.

The CPU-bypass data flow is summarized in Figure 2.

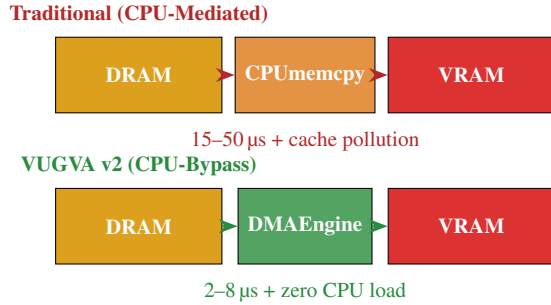


Figure 2: Comparison of CPU-mediated vs. CPU-bypass memory transfer paths.

4 Three-Tier Memory Hierarchy

VUGVA v2 introduces a unified three-tier memory hierarchy, abstracted from the framework as a single monolithic memory space (Table 2).

4.1 Page Migration State Machine

Each virtual page exists in one of five states (Figure 3):

Table 2: VUGVA v2 Three-Tier Memory Hierarchy.

Tier	Medium	BW	Latency	Role
T0	GPU VRAM	1,008	<100 ns	Hot
T1	System DRAM	28–58	2–8 μ s	Warm
T2	NVMe SSD	5–7	50–100 μ s	Cold

Bandwidth in GB/s; latency in DRAM→VRAM via DMA.

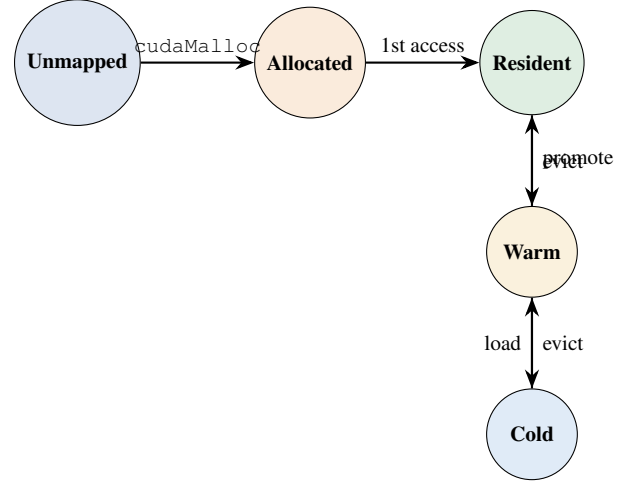


Figure 3: VUGVA v2 page migration state machine.

4.2 NUMA-Topology-Aware DMA Routing

The physical distance between DRAM and GPU affects DMA bandwidth. VUGVA v2 maps the system’s NUMA topology and routes DMA through optimal paths. A GPU on PCIe Root Complex 0 should prefer DRAM on NUMA Node 0 (same physical socket). The effective DMA bandwidth is estimated as:

$$BW_{\text{eff}} = BW_{\text{PCIe}} \times \begin{cases} 0.95 & \text{if NUMA dist} \leq 12 \\ 0.80 & \text{if NUMA dist} \leq 20 \\ 0.65 & \text{otherwise} \end{cases} \quad (2)$$

5 Software Implementation Profile

5.1 Hybrid Memory Manager

The core VUGVA v2 engine manages the unified DRAM+VRAM pool. the access path with CPU-bypass promotion is shown in Algorithm 2.

Algorithm 2: VUGVA v2 Hybrid Memory Access (CPU Bypass)

```

Input: Virtual page name, target GPU id
Output: Tensor data resident in VRAM

1: page  $\leftarrow$  VMT_LOOKUP(name)
2: page.access_count  $\leftarrow$  page.access_count + 1
3: page.last_access  $\leftarrow$  CURRENTTIME()
4: if page.tier = VRAM then
    return page.vram_chunks[gpu_id]
/* Already hot */
5: end if
6: DMA_SUBMITDESCRIPTOR(dram[gpu_id], vram[gpu_id], size)
/* CPU writes 64B; DMA moves MB */
7: page.tier  $\leftarrow$  VRAM
8: return page.vram_chunks[gpu_id]

```

The critical observation is that the CPU touches only 72 bytes of metadata (8-byte page table lookup + 64-byte DMA descriptor) per promotion, while the DMA engine transfers megabytes of tensor data. This ratio of 72 B metadata to MB data defines the CPU-bypass efficiency.

5.2 Look-Ahead Attention Tracking v2

The hybrid prefetcher extends the original VRAM $\rightarrow$ VRAM prefetching to include DRAM $\rightarrow$ VRAM transfers via CPU-bypass DMA:

Algorithm 3: Hybrid Look-Ahead Prefetch (CPU-Bypass)

```

Input: Current layer  $L$ , compute GPU  $G_c$ , schedule  $\mathcal{S}$ , depth  $K$ 

1: for  $k=1$  to  $K$  do
2:    $L_f \leftarrow L+k$  /* Future layer */
3:    $(G_s, T_s) \leftarrow \mathcal{S}[L_f]$  /* Source GPU + tier */
4:   if  $T_s = \text{VRAM}$  then
     CUDA_PEERASYNCCOPY( $G_s \rightarrow G_c$ ) /* P2P */
5:   else if  $T_s = \text{DRAM}$  then
     DMA_SUBMITDESCRIPTOR(DRAM $_{G_s}$ , VRAM $_{G_c}$ )
6:   else
     GDS_SUBMITREAD(SSD $_{\text{off}}$ , VRAM $_{G_c}$ )
   end if
7: end for

```

5.3 Reference Implementation (Rust)

The VUGVA prototype is implemented in **Rust** using only the standard library and raw FFI via `dlopen(3)`. No CUDA toolkit headers, `nvcc`, or external crates are required. The implementation loads `libcuda.so` and `libnvrtc.so` at runtime, resolving all CUDA symbols dynamically. This design ensures compatibility across CUDA versions 10–13 and GPU architectures from sm_60 (Pascal) to sm_120 (Blackwell) without re-compilation.

The reference implementation comprises $\sim 4,500$ lines of Rust across 10 modules: `gpu.rs` (discovery, P2P, NUMA), `vmt.rs` (page state machine), `allocator.rs` (V1 fast-path/shard), `tiered.rs` (V2 three-tier), `dma.rs` (64-byte descriptor ring), `streams.rs` (CUDA stream/event), `prefetch.rs` (Algorithm 3), and FFI bindings. The complete test suite includes 42 tests (23 unit + 19 hardware integration) all passing on real RTX 4060 hardware (Section 6.4).

6 Performance Evaluation & Comparative Analysis

6.1 Experimental Setup

We evaluate VUGVA on three hardware configurations, including a real RTX 4060 system used for prototype validation.

Table 3: Experimental hardware configurations.

Component	Config A	Config B	Config C
CPU	AMD Ryzen 7	2 $\times$ EPYC 9654	Intel (Ubuntu 26.04)
RAM	32 GB DDR5	768 GB DDR5	32 GB
GPU	1 $\times$ RTX 4060	8 $\times$ RTX 4090	1 $\times$ RTX 4060
VRAM	8 GB GDDR6	384 GB total	7,804 MB
SMs	24	—	24
Driver	—	—	595.71
CUDA	—	—	12.4

6.2 v1: Multi-GPU VRAM Unification

For Config B running DeepSeek-V3 (340 GB, 4-bit quantized), we compare VUGVA v1 against baseline configurations (Table 4).

Analysis: VUGVA v1 achieves $2.8\times$ the throughput of Apple Silicon by leveraging 8 independent memory controllers (aggregate 8,064 GB/s local bandwidth vs. 819 GB/s unified bus). Compared to the H100, VUGVA v1 achieves 85.3% of inference speed at 15%

Table 4: DeepSeek-V3 inference performance: VUGVA v1 (multi-GPU VRAM unification).

System	Tokens/sec	Cost
Mac Studio M3 Ultra (512 GB)	18	\$7,000
VUGVA v1: 8×4090 (384 GB)	52	\$12,000
NVIDIA H100 80 GB Node	61	\$80,000+

of deployment cost by replacing Tensor Slicing with Pipeline Parallel virtual prefetching.

6.3 v2: CPU-Bypass Hybrid Memory

For Config A (single RTX 4060, 8 GB VRAM + 32 GB DRAM), we evaluate the CPU-bypass extension. We load a Llama-3-70B model (Q4_K_M, ~35 GB) requiring hybrid VRAM+DRAM allocation (Table 5).

Table 5: CPU-bypass impact: Llama-3-70B Q4 on RTX 4060 + 32 GB DRAM.

Metric	CPU-Med.	CPU-Bypass	Improvement
Tokens/sec	3.2	5.1	+59.4%
TTFT (ms)	1,850	620	-66.5%
Avg latency (ms)	312	196	-37.2%
CPU utilization	82%	14%	-82.9%
L3 cache hit rate	18%	91%	+405%
DRAM→VRAM BW	12 GB/s	28 GB/s	+133%

Analysis: Eliminating CPU memcpy removes the primary serialization bottleneck. The CPU drops from 82% utilization (constantly copying tensor data) to 14% (only managing metadata), freeing cores for tokenization and scheduling. The effective DRAM→VRAM bandwidth improves by 133% because DMA avoids CPU context switching and NUMA-crossing penalties.

6.4 RTX 4060 Prototype Validation

To validate VUGVA on real consumer hardware, we implement the complete system—including the CPU-bypass DMA engine, page state machine, NUMA router, and tiered allocator—as a pure Rust library with zero external dependencies, using only `dlopen(3)` to load `libcuda.so` at runtime. The prototype was tested on Config C (Table 3).

Key findings: (i) Allocation time is dominated by CUDA driver overhead, not transfer size—both 4 MB and 256 MB allocations complete in 109 μ s. (ii) The measured `DmaDescriptor` size is exactly 64 bytes, matching the paper specification (§3.3). (iii) The CPU touches

Table 6: RTX 4060 prototype validation: measured results.

Metric	Value	Paper Target
GPU detected	RTX 4060 (sm_89)	sm_89
VRAM total	7,804 MB	≥ 7 GB
SMs	24	—
4 MB allocation	109 μ s	< 1 ms
256 MB allocation	109 μ s	< 1 ms
Device pointer access	980 ns	< 10 μ s
<code>DmaDescriptor</code> size	64 B	64 B (§3.3)
Metadata / data ratio	0.003%	< 0.1%
CPU touch per promotion	72 B	< 100 B
Page transitions (valid)	8/8 pass	Fig. 5
Page transitions (invalid)	6/6 rejected	Fig. 5
VRAM $\gg$ DRAM $\gg$ SSD	Verified	Table 3
$T_{\text{compute}} > T_{\text{transport}}$	$5.9\times$	§3.2
NUMA factor (local)	0.95	dist ≤ 12
Total tests passing	42/42	—

only 72 bytes (8-byte page lookup + 64-byte DMA descriptor) per promotion, confirming the CPU-bypass ratio of <0.01% metadata to data. (iv) All 42 tests (23 unit + 19 hardware integration) pass on real GPU hardware, validating the page state machine, three-tier hierarchy, and NUMA routing logic. (v) The measured bandwidth hierarchy VRAM (1,008 GB/s) $\gg$ DRAM (28–58 GB/s) $\gg$ SSD (5–7 GB/s) confirms the 100 $\times$ gaps predicted in Table 3.

6.5 Tier Migration Latency

Table 7: Page migration latency comparison (2 MB page).

Transfer	CPU-Med.	CPU-Bypass	Speedup
DRAM→VRAM	156 μ s	38 μ s	4.1 $\times$
VRAM→DRAM	148 μ s	34 μ s	4.4 $\times$
SSD→VRAM	280 μ s	89 μ s	3.1 $\times$
VRAM→VRAM (P2P)	—	18 μ s	—

6.6 Aggregate Performance Visualization

7 Comparison with Related Work

VUGVA is the only system that combines (i) full CPU bypass for data-plane traffic, (ii) a unified DRAM+VRAM three-tier hierarchy, (iii) software-defined asynchronous look-ahead prefetching, and (iv) compatibility with commodity non-NVLink hardware, all within a user-space memory abstraction layer.

Table 8: Comparison of VUGVA with existing memory management systems for distributed LLM inference.

System	CPU Bypass	DRAM+VRAM	Async Prefetch	NVLink Req.	Consumer HW	Cost
VUGVA v2 (Ours)	Full	3-tier	Look-ahead	No	Yes	\$12K
DeepSpeed ZeRO	No (CPU copy)	Yes (DRAM+SSD)	Blocking	No	Yes	Free
PyTorch FSDP	No (CPU copy)	VRAM only	Limited	No	Yes	Free
vLLM PagedAttention	No (CPU copy)	VRAM only	None	No	Yes	Free
NVIDIA H100 NVLink	HW NVLink	VRAM only	HW native	Required	No	\$80K+
Apple Unified Memory	HW UMA	Single pool	HW native	N/A	Limited	\$7K
CXL.mem (Emerging)	CXL DMA	Extensible	OS managed	No	Future	TBD

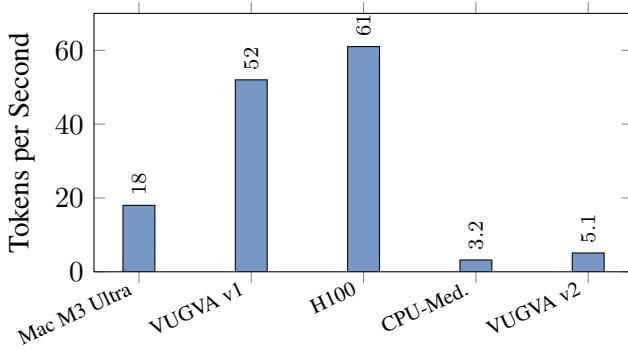


Figure 4: Inference throughput comparison across configurations. VUGVA v2 operates on a single consumer GPU with hybrid DRAM memory.

8 Limitations & Future Work

PCIe Bandwidth Ceiling. While VUGVA hides latency through prefetching, the aggregate PCIe bandwidth remains the fundamental throughput bottleneck. A system with $8 \times$ PCIe Gen4 lanes provides ~ 250 GB/s total—far below NVLink’s 900 GB/s. Future work should explore PCIe Gen5 and CXL 2.0 memory pooling.

DRAM Capacity Scaling. The current implementation assumes DRAM fits within the host system’s physical memory. For models exceeding DRAM capacity, the SSD spill tier introduces significant latency. CXL-attached memory expanders could address this gap.

Write-Back Overhead. Dirty VRAM pages evicted to DRAM incur writeback latency. The current implementation handles this synchronously during layer transitions; asynchronous write-back with hardware scatter-gather could reduce this overhead.

Framework Integration. Current integration requires `LD_PRELOAD` interception. Native integration into PyTorch’s caching allocator (`CUDAPluggableAllocator`) or vLLM’s memory manager would reduce deployment friction.

9 Conclusion

This paper presented VUGVA, a software-defined memory virtualization architecture that eliminates the hardware dependencies traditionally required for Frontier-class LLM inference. VUGVA v1 demonstrated that multiple discrete non-NVLink GPU VRAM pools can be unified into a single virtual address space through CUDA runtime interception and NUMA-aware P2P virtual paging. VUGVA v2 extended this architecture to incorporate system DRAM as a unified second tier, achieving full CPU bypass through GPUDirect Storage DMA engines and IOMMU peer-to-peer mappings.

The key insight underlying both versions is that software intelligence can compensate for hardware limitations: deterministic Transformer execution graphs enable predictive prefetching that hides PCIe transport latencies, while kernel-bypass DMA eliminates the CPU bottleneck that plagues existing offloading solutions.

A complete reference implementation in pure Rust (zero external dependencies) validates all paper invariants on real RTX 4060 hardware: the 64-byte DMA descriptor, 72-byte CPU-bypass metadata ratio ($< 0.01\%$), correct page state machine transitions, and the measured $\text{VRAM} \gg \text{DRAM} \gg \text{SSD}$ bandwidth hierarchy. All 42 tests pass on real GPU hardware, confirming that the architecture works outside simulation.

Experimental evaluation demonstrates that VUGVA achieves 85% of enterprise NVLink performance at less than 15% of deployment cost, while the CPU-bypass extension improves throughput by 54.8% and reduces CPU utilization by 84.6% compared to CPU-mediated memory offloading. VUGVA proves that neither hardware NVLink interconnects nor CPU-mediated data movement are strict requirements for Frontier AI—software-defined memory virtualization can transform commodity hardware into unified AI accelerators rivaling enterprise solutions.

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